

YICHEN (ANNA) TU
+1 (984)317-9902 | anna.tu@duke.edu | Durham, NC

EDUCATION

Duke University, Pratt School of Engineering , Durham, NC	Expected May 2027
Master of Engineering (M.Eng.), Financial Technology	
Relevant Coursework: Programming (C++, Python), Software Engineering (C++, Python), Asset Pricing, Machine Learning, Quantitative Risk Management	
University of St Andrews , St Andrews, Scotland	Sep 2021 – May 2025
Bachelor of Science (Honours) in Economics and Mathematics	
Relevant Coursework: Causal Inference, Time Series Analysis, Mathematical Modeling, Asset Pricing, Advanced Linear Algebra	
University of Virginia (Exchange Student), Charlottesville, VA	Aug 2023 – May 2024
Relevant Coursework: Real and Complex Analysis, Econometrics, Stochastic Processes Technical: C++, Python (pandas, numpy, scikit-learn), R Studio, SQL, MATLAB, Latex	

PROFESSIONAL EXPERIENCE

Fortune SG Fund Management	Shanghai, China
<i>Global Markets Intern (Investment Department)</i>	Jun 2025 – Aug 2025
- Collected, cleaned, and automated large datasets on daily NAV, valuation ratios, and sector exposures using Excel/Python; built templates that reduced manual monitoring time by ~30% and delivered more consistent portfolio tracking for senior analysts - Calculated risk factors, including volatility, beta sensitivity, and concentration exposures, for multiple ETFs and mutual funds, supported stress-testing scenarios that highlighted downside exposures and informed investment committee meetings discussions - Performed bottom-up and macro-linked analysis on equity and energy funds (e.g., ARKF, HuaAn Oil & Gas LOF), connecting with OPEC+ output and market news, and synthesized findings into fund presentations and investor roadshow materials	
Hwabao Securities	Shanghai, China
<i>Mutual Fund Research Intern (Research and Innovation Department)</i>	Jun 2024 – Aug 2024
- Processed stock weights and factor exposure datasets across 9 indices (~1,000 data points per stock) using Python, built the foundation for a multi-factor risk model, supported portfolio exposure analysis and visualized key insights - Validated mutual fund and index datasets, applied Pandas to compute daily PnL, historical VaR, and exposure breakdowns; designed Excel dashboards with charts, pivot tables, and slicers to track portfolio risk metrics and present findings to senior - Analyzed ~50,000 raw data points using multivariate regression to identify operational risk factors that significantly influenced process time—such as returned data and stage-specific approval differences—and assisted in streamlining process management	

PROJECTS AND RESEARCH EXPERIENCE

Volatility Forecasting on QQQ (Python)	Jan 2026 – May 2026
- Modeled conditional volatility using EWMA, GARCH(1,1), and GJR-GARCH(1,1) on 26 years of QQQ daily returns; GJR-GARCH reduced out-of-sample MSE by ~8% vs. EWMA over a 1,323-observation expanding window. - Backtested VaR and Expected Shortfall at 95%/99% confidence levels; GJR-GARCH achieved lowest exceedance rate (5.74%) and highest Sharpe-like ratio (0.052) under a volatility-targeting strategy.	
Standardization and IPW for Causal Inference - Honours Dissertation (R Studio)	Sep 2024 – May 2025
- Estimated public health risk of smoking on overweight incidence using BRFSS 2023 data (~400,000 obs); implemented both standardization and inverse probability weighting (IPW) methods in R Studio to assess robustness across modeling assumptions - Found ATE estimates ranging from -5.5% (IPW) to -9.3% (standardization); highlighted robustness in risk quantification	
Research Assistant - Business School (Python)	Sep 2024 – Dec 2024
- Built a scalable Python data pipeline to link LinkedIn employment records, patent inventor, and assignee datasets via fuzzy matching and probabilistic record linkage - Improved entity-resolution accuracy by 5–17% across 1TB+ of data, enabling faculty research on corporate innovation.	
STEP Diving into Data 2024 (R Studio)	May 2024 – Jul 2024
- Directed a 7-member team in a university data challenge analyzing 3,000+ ADHD case records and presented to mentors - Conducted EDA and preprocessing in R (data merging, missing-value handling, numeric parsing, categorical encoding) - Found measurable associations between soil contamination, heavy metal exposure, and ADHD incidence	